

Batch size	CPU (s)	GPU (s)	Speedup (CPU/GPU)	Loss
64	408.458	30.7409	13.28711911	0.0352917
128	427.667	17.7144	24.14233618	0.15328
256	440.341	10.3596	42.50559867	0.21985
512	456.595	8.13008	56.16119399	0.266631
1024	598.016	6.7978	87.97199094	0.310848
2048	747.898	5.64219	132.5545577	0.416119
4096	864.14	5.98207	144.4550131	0.591687
8192	746.529	5.65745	131.9550327	0.831569
16384	1031.2	5.71409	180.466181	0.952762
32,768	793.577	4.0845	194.2898764	1.08545
50000	1275.13	6.47579	196.9072499	1.08834
Batch size PyTorch time (s) Final avg loss				

64	9.2563	0.020042		
128	4.7769	0.144012		
256	2.2301	0.183786		
512	1.1093	0.247421		
1024	0.5471	0.299811		
2048	0.2670	0.359874		
4096	0.1493	0.547881		
8192	0.1001	0.669435		
16384	0.0952	0.958462		
32768	0.0605	1.072812		
50000	0.0894	1.055113		